

① Heuristic design & admissibility

Steps

```

count = 0
for each position i:
    if s[i] != goal[i]:
        count += 1

```

⇒ This is a lower bound because every action changes one position, so the heuristic lower-bounds the minimum number of letter replacements required to reach the goal.

⇒ It can't overestimate because each action has a cost of exactly 1, so the true remaining cost must be at least $d(\text{current word}, \text{goal word})$
 $d(s, g)$

Scrabble

```

cost = 0
for each position i:
    if s[i] != goal[i]:
        cost += scrabble_val(goal[i])

```

s_i = letter from current word
 g_i = goal letter

⇒ For every position where $s_i \neq g_i$, that position must eventually be changed. The final replacement at that position must insert the goal letter.

⇒ Every differing position must eventually be changed, and changing that position to its required goal letter increases Scrabble value. Replacements will have costs, so they cannot make the actual optimal cost less than the sum of the replacement costs.

Frequency

```

count = 0
for each position i:
    if s[i] != goal[i]:
        count += 1

```

⇒ Lower-bounds the minimum frequency cost of the remaining actions. If there are $d(s, g)$ mismatched positions, at least $d(s, g)$ actions are required, and action costs at least 1.

⇒ Every remaining action costs at least 1, and at least $d(s, g)$ actions are necessary, so it can't overestimate the true optimal remaining frequency cost.

Question 2: Does the heuristic actually help? [10 points]

Run both **uniform-cost search** and **A*** on:

- WHY → ASK
- BEAR → DUCK

Run each problem using all three cost measures. Report:

Word pair	Cost measure	UCS cost	UCS nodes expanded	A* cost	A* nodes expanded	Reduction
WHY → ASK	Steps	5	773	5	16	97.93%
WHY → ASK	Scrabble	9	765	9	13	98.30%
WHY → ASK	Frequency	38.85540652	808	38.85540652	755	6.56%
BEAR → DUCK	Steps	4	691	4	6	99.13%
BEAR → DUCK	Scrabble	11	2981	11	6	99.80%
BEAR → DUCK	Frequency	38.2402284	1084	38.2402284	720	33.58%

Calculate the percentage reduction as:

$$100 \times \frac{\text{UCS expansions} - \text{A* expansions}}{\text{UCS expansions}}$$

(2)

1. UCS & A* always return the same optimal cost.

2. Scrabble produced the largest reduction for WHY $\rightarrow$ ASK

Scrabble produced the largest reduction for BEAR $\rightarrow$ DUCK

3. Frequency heuristic was least informative for WHY $\rightarrow$ ASK

Frequency heuristic was least informative for BEAR $\rightarrow$ DUCK

4. Steps heuristic: Since every action changes exactly one position, this provides a lower bound on the remaining actions making it informative when the number of required changes closely represents the actual search cost.

Scrabble heuristic is more informative for the Scrabble cost because it adds the scrabble values of the goal letters that still need to be inserted.

Frequency heuristic was less informative because it only counts the number of mismatched positions. It ignores the rarity of the words produced along the path and every frequency action costs $1 + \text{rarity}(w)$, so the actual cost can be much larger than the heuristic estimate.

(3) optimal cost $w_1 + w_2 = \text{optimal cost } w_2 + w_1$

STEPS: Yes because every action costs exactly 1, since the actions are reversible and their costs are symmetric.

SCRABBLE: $\text{cost}(\text{cat}, \text{cab}) = 3$ and $\text{cost}(\text{cab}, \text{cat}) = 1$

No because the costs are not symmetric.

FREQUENCY: No because the costs can be different in opposite directions.

$$1 + \text{rarity}(\text{beat}) \neq 1 + \text{rarity}(\text{bear})$$